25 (a) Write 4.72×10^6 as an ordinary number.

$4.72 \times 10^6 = 4\ 720\ 000$

1 mark
Move each digit six places left.

4 720 000

(1)

(b) Write 0.0023 in standard form.

$0.0023 = 2.3 \times 10^{-3}$

1 mark
Use $1 \leq a < 10$.

2.3×10^{-3}

(1)

(c) Show that $(3 \times 10^2)^3 > 20$ million

$(3 \times 10^2)^3 = 3^3 \times (10^2)^3$

$= 27 \times 10^6 = 27\ 000\ 000$

1 mark
Cube BOTH factors.

$27\ 000\ 000 > 20\ 000\ 000$, as required.

1 mark
Compare the two values explicitly.

(2)

(Total for Question 25 is 4 marks)

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